Ramond superconformal blocks from two Virasoro algebras

Abstract

We derive the precise consequence of the embedding $\mathrm{Vir}\oplus\mathrm{Vir}\subset\mathrm{F}\oplus\mathrm{SVir}$ for the mixed NS–R–R genus-two theta block. The embedding is not an isomorphism of chiral algebras. One must first tensor the super-Virasoro block with a block of an auxiliary Majorana fermion. The resulting block decomposes into a sum of products of two ordinary Virasoro blocks. We translate the two Virasoro central charges and all internal weights to the conventions of the central-charge recursion notes, prove the NS and Ramond branching grades, and retain the two Ramond parity copies. We define the two oriented branching coefficients by restriction of the NS–R–R three-point forms. The Grassmann signs at the two vertices and in the three propagators are then derived separately. This gives an exact sewing identity. We also give the finite-product candidate for the momentum dependence of the branching coefficients. Its all-NS specialization is proved, but its mixed Ramond specialization is conditional on the proposed Ramond double-Liouville correspondence and on an unproved Ramond one-leg normalization, supported by the conjectured Ramond Whittaker coefficient. The exact identity is not conditional on this candidate.

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1 Statement and conventions

1.1 Conventions

We use the ordinary central charge and the super-Virasoro algebra

$$\begin{aligned}
[L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12}(m^3-m)\delta_{m+n,0}, \\
[L_m, G_r] &= \left(\frac{m}{2} - r\right) G_{m+r}, \\
\{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3}\left(r^2 - \frac{1}{4}\right)\delta_{r+s,0}.
\end{aligned} \tag{1.1}$$

The fermionic indices are half-integral in the NS sector and integral in the Ramond sector. As in the main notes, we write

$$Q = b + b^{-1}, \qquad c = \frac{3}{2} + 3Q^2. \tag{1.2}$$

The NS weight on the first edge of the theta graph is parameterized by

$$h_1 = \frac{Q^2 - \lambda_1^2}{8}. \tag{1.3}$$

The two Ramond weights are

$$h_i = \frac{c}{24} - \beta_i^2, \qquad i = 2, 3. \tag{1.4}$$

The Ramond ground states and their bilinear BPZ form obey

$$\begin{aligned} G_0 w_i^{\pm} &= i\beta_i e^{\mp i\pi/4} w_i^{\mp}, & \langle w_i^+ | w_i^+ \rangle &= 1, \\ \langle w_i^- | w_i^- \rangle &= i, & \langle w_i^+ | w_i^- \rangle &= 0. \end{aligned} \tag{1.5}$$

In particular, $G_0^2 = -\beta_i^2 = L_0 - c/24$ on the ground states.

The superconformal block of interest is the stripped mixed theta block

$$\mathbb{F}_f^{(\eta,\eta')}(h_1, \beta_2, \beta_3, c), \qquad f = 0, 1, \qquad \eta, \eta' = \pm 1, \tag{1.6}$$

with slot order

$$(\mathrm{NS}, \mathrm{R}, \mathrm{R}) = (\infty, 1, 0). \tag{1.7}$$

The plumbing parameters are (q_i, η_i), with $\eta_i = \pm 1$. The labels η, η' select the two NS–R–R chiral structures at the two vertices, whereas the η_i are the lifts used in the three plumbing tubes. These two uses of the letter η are the same as in the main notes and will not be interchanged.

1.2 Results

The result is not an equality between $\mathbb{F}_f^{(\eta,\eta')}$ and one ordinary Virasoro block. One must first multiply by the known block $\mathbb{F}_g^{\mathrm{f}}$ of an auxiliary Majorana fermion. With the definitions given below, the exact identity is

$$\boxed{\begin{aligned} &\mathbb{F}_f^{(\eta,\eta')}(h_1, \beta_2, \beta_3, c)\mathbb{F}_g^{\mathrm{f}} \\ &= \sum_{\substack{n_1 \in \frac{1}{2}\mathbb{Z}, \quad n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ \epsilon_2, \epsilon_3 \in \{0,1\} \\ 2n_1 + \epsilon_2 + \epsilon_3 = f + g \pmod 2}} q_1^{2n_1^2} q_2^{2n_2^2 - 1/8} q_3^{2n_3^2 - 1/8} \eta_1^{2n_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3} \\ &\quad \times (-1)^{(2n_1)\epsilon_2 + (2n_1)\epsilon_3 + \epsilon_2\epsilon_3} B_{f,g}^{(\eta),L}(n_1, n_2, n_3; \epsilon_2, \epsilon_3) B_{f,g}^{(\eta'),R}(n_1, n_2, n_3; \epsilon_2, \epsilon_3) \\ &\quad \times \prod_{a=1}^{2} F_\theta\left(c^{(a)}; h_{1,n_1}^{(a)}, h_{2,n_2}^{(a)}, h_{3,n_3}^{(a)}\right). \end{aligned}} \tag{1.8}$$

The two branching coefficients are the restrictions of the two oriented three-point forms to the branch highest states. No relation between the left and right coefficients is assumed. Their proposed finite-product form is

$$\boxed{B^{(\eta),L}_{f,g} = \kappa^{(\eta),L}_{f,g}\mathcal{C}(n_1,n_2,n_3), \qquad B^{(\eta'),R}_{f,g} = \kappa^{(\eta'),R}_{f,g}\mathcal{C}(n_1,n_2,n_3).} \tag{1.9}$$

The arguments $(n_1, n_2, n_3; \epsilon_2, \epsilon_3)$ of B and κ are suppressed only in Eq. (1.9). The explicit finite product $\mathcal{C}$ is given in Eq. (5.5). It is proved when all three modules are NS. Its use in the mixed NS–R–R identity (1.8) is conditional. The exact identity itself uses the coefficients as defined by the three-point forms and does not depend on this finite-product proposal.

2 The two commuting Virasoro algebras

2.1 The auxiliary fermion

Introduce one Majorana fermion with modes f_r satisfying

$$\{f_r, f_s\} = \delta_{r+s,0}, \qquad \{f_r, G_s\} = 0, \qquad [f_r, L_n] = 0. \tag{2.1}$$

The f_r have the same moding as the G_r. Thus the auxiliary fermion is in its NS sector on an NS edge and in its Ramond sector on a Ramond edge. Its stress tensor has modes

$$L^{\mathrm{f}}_n = \frac{1}{2}\sum_{r\in\mathbb{Z}+\delta} r : f_{n-r}f_r : + \frac{1-2\delta}{16}\delta_{n,0}, \qquad \delta = \begin{cases} \frac{1}{2}, & \text{NS}, \\ 0, & \text{R}. \end{cases} \tag{2.2}$$

They satisfy a Virasoro algebra of central charge 1/2 and

$$[L^{\mathrm{f}}_n, f_r] = \left(-\frac{n}{2} - r\right) f_{n+r}. \tag{2.3}$$

We use

$$f_r^\dagger = -f_{-r}. \tag{2.4}$$

In the Ramond sector let $u^\pm$ be even and odd ground states with

$$f_0 u^\pm = \frac{1}{\sqrt{2}}u^\mp, \quad \langle u^+|u^+\rangle = 1, \quad \langle u^-|u^-\rangle = -1, \tag{2.5}$$

and vanishing mixed inner products. Equation (2.4) forces the relative minus sign between the two norms. Both ground states have fermion weight 1/16.

It is convenient to put the auxiliary fermion first in the tensor product. Thus f_r acts on the first factor, while an odd super-Virasoro operator acts with the parity sign of that factor. Explicitly,

$$G_r(u \otimes x) = (-1)^{|u|} u \otimes G_r x. \tag{2.6}$$

This convention makes f_r and G_s anticommute, as required by Eq. (2.1). The bilinear form on the tensor product is the product of the two bilinear forms. No complex conjugation of coefficients is made.

2.2 Definition of the two stress tensors

Define the even modes

$$U_n = \sum_{r \in \mathbb{Z}+\delta} f_{n-r} G_r. \tag{2.7}$$

For $b^2 \neq 1$, set

$$\begin{aligned} L_n^{(1)} &= \frac{b^{-1}}{b^{-1}-b} L_n - \frac{b^{-1}+2b}{b^{-1}-b} L_n^{\mathrm{f}} + \frac{1}{b^{-1}-b} U_n, \\ L_n^{(2)} &= \frac{b}{b-b^{-1}} L_n - \frac{b+2b^{-1}}{b-b^{-1}} L_n^{\mathrm{f}} + \frac{1}{b-b^{-1}} U_n. \end{aligned} \tag{2.8}$$

The sums in U_n are well defined on every vector in a highest-weight module because only finitely many terms act nontrivially on that vector. The BPZ rules give

$$(L_n^{\mathrm{f}})^\dagger = L_{-n}^{\mathrm{f}}, \qquad U_n^\dagger = U_{-n}, \qquad (L_n^{(i)})^\dagger = L_{-n}^{(i)}. \tag{2.9}$$

For the middle equality, reversing $f_{n-r}G_r$ first produces a minus sign from Eq. (2.4); moving G_{-r} back through the fermion produces a second minus sign. Thus the embedded Virasoro Gram matrices use the same bilinear BPZ convention as the original block.

We now verify rather than assume that Eq. (2.8) gives two commuting Virasoro algebras. Let T, T^{f}, U be the fields with modes $L_n, L_n^{\mathrm{f}}, U_n$. Wick contraction using Eqs. (1.1) and (2.1) gives the following singular products:

$$
\begin{aligned}
T(z)T(w) &\sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \\
T^{\mathrm{f}}(z)T^{\mathrm{f}}(w) &\sim \frac{1/4}{(z-w)^4} + \frac{2T^{\mathrm{f}}(w)}{(z-w)^2} + \frac{\partial T^{\mathrm{f}}(w)}{z-w}, \\
T(z)U(w) + U(z)T(w) &\sim \frac{3U(w)}{(z-w)^2} + \frac{3\partial U(w)/2}{z-w}, \\
T^{\mathrm{f}}(z)U(w) + U(z)T^{\mathrm{f}}(w) &\sim \frac{U(w)}{(z-w)^2} + \frac{\partial U(w)/2}{z-w}, \\
U(z)U(w) &\sim -\frac{2c/3}{(z-w)^4} - \frac{2T(w) + (4c/3)T^{\mathrm{f}}(w)}{(z-w)^2} - \frac{\partial T(w) + (2c/3)\partial T^{\mathrm{f}}(w)}{z-w}.
\end{aligned}
\tag{2.10}
$$

For example, the fourth-order pole in the last line is the double contraction of $f(z)$ with $f(w)$ and $G(z)$ with $G(w)$. Moving the odd $G(z)$ through the odd $f(w)$ gives the displayed minus sign. The second-order T^{f} term follows by expanding $:f(z)f(w):= 2(z-w)T^{\mathrm{f}}(w) + O((z-w)^2)$. The first-order terms are one half of the derivatives of the second-order coefficients, as required by exchanging z and w.

Consider a linear combination

$$
\mathcal{T} = aT + dT^{\mathrm{f}} + eU. \tag{2.11}
$$

Reading the second- and first-order poles from Eq. (2.10), its self-OPE is a Virasoro OPE precisely when

$$
a^2 - e^2 = a, \quad d^2 - \frac{2c}{3}e^2 = d, \quad 3a + d = 2. \tag{2.12}
$$

The corresponding central charge is

$$
c_{\mathcal{T}} = a^2 c + \frac{d^2}{2} - \frac{4c}{3}e^2. \tag{2.13}
$$

For the two rows of coefficients in Eq. (2.8), direct substitution into Eq. (2.12) gives zero on the left after moving the right side to the left. The mixed OPE vanishes because the same substitution gives

$$
\begin{aligned}
a_1a_2 - e_1e_2 &= 0, \\
d_1d_2 - \frac{2c}{3}e_1e_2 &= 0, \\
3(a_1e_2 + a_2e_1) + (d_1e_2 + d_2e_1) &= 0, \\
a_1a_2c + \frac{d_1d_2}{2} - \frac{4c}{3}e_1e_2 &= 0.
\end{aligned}
\tag{2.14}
$$

The first three equations remove the second- and first-order poles and the last removes the fourth-order pole. Hence

$$[L_n^{(1)}, L_m^{(2)}] = 0. \tag{2.15}$$

This proves the algebra embedding.

The sums of the three coefficients in the two rows are respectively

$$a_1 + a_2 = 1, \qquad d_1 + d_2 = 1, \qquad e_1 + e_2 = 0. \tag{2.16}$$

Therefore the total stress tensor is unchanged:

$$L_n^{(1)} + L_n^{(2)} = L_n + L_n^{\mathrm{f}}. \tag{2.17}$$

This identity will determine the plumbing grades in the branching decomposition.

2.3 The two central charges

Introduce $b^{(1)}$ and $b^{(2)}$ by

$$(b^{(1)})^2 = \frac{2b^2}{1-b^2}, \qquad (b^{(2)})^{-2} = \frac{2b^{-2}}{1-b^{-2}} = \frac{2}{b^2-1}. \tag{2.18}$$

Set

$$Q^{(i)} = b^{(i)} + (b^{(i)})^{-1}, \qquad c^{(i)} = 1 + 6(Q^{(i)})^2. \tag{2.19}$$

Equation (2.13) applied to Eq. (2.8) gives precisely these two values. An elementary calculation from Eq. (2.18) gives

$$(Q^{(1)})^2 + (Q^{(2)})^2 = \frac{Q^2}{2}. \tag{2.20}$$

It follows that

$$c^{(1)} + c^{(2)} = 2 + 3Q^2 = c + \frac{1}{2}. \tag{2.21}$$

The additional 1/2 is exactly the central charge of the auxiliary fermion. Notice also that Eq. (2.8), rather than an equality of the three chiral algebras, proves only the inclusion

$$\mathrm{Vir}_{c^{(1)}} \oplus \mathrm{Vir}_{c^{(2)}} \subset \mathrm{F} \oplus \mathrm{SVir}_c. \tag{2.22}$$

3 Branching of the NS and Ramond modules

3.1 The Virasoro weights

It is useful to use one momentum for each of the three internal edges:

$$P_1 = \frac{\lambda_1}{2}, \qquad P_2 = \sqrt{2}\,\beta_2, \qquad P_3 = \sqrt{2}\,\beta_3. \tag{3.1}$$

For every allowed branching label n, define

$$\begin{aligned} P_{i,n}^{(1)} &= \frac{P_i}{\sqrt{2-2b^2}} + nb^{(1)}, \\ P_{i,n}^{(2)} &= \frac{P_i}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}, \end{aligned} \tag{3.2}$$

and

$$h_{i,n}^{(a)} = \frac{(Q^{(a)})^2}{4} - (P_{i,n}^{(a)})^2, \qquad a = 1, 2. \tag{3.3}$$

The square-root branches in Eqs. (2.18) and (3.2) are chosen consistently. They can and will be chosen so that

$$\begin{aligned} \frac{b^{(1)}}{\sqrt{2-2b^2}} + \frac{(b^{(2)})^{-1}}{\sqrt{2-2b^{-2}}} &= 0, \\ \frac{1}{2-2b^2} + \frac{1}{2-2b^{-2}} &= \frac{1}{2}, \\ (b^{(1)})^2 + (b^{(2)})^{-2} &= -2. \end{aligned} \tag{3.4}$$

Using Eqs. (2.20) and (3.4) in Eq. (3.3) gives, one term at a time,

$$\begin{aligned} h_{i,n}^{(1)} + h_{i,n}^{(2)} &= \frac{Q^2}{8} - P_i^2 \left(\frac{1}{2-2b^2} + \frac{1}{2-2b^{-2}} \right) \\ &\quad - 2nP_i \left(\frac{b^{(1)}}{\sqrt{2-2b^2}} + \frac{(b^{(2)})^{-1}}{\sqrt{2-2b^{-2}}} \right) \\ &\quad - n^2 \big((b^{(1)})^2 + (b^{(2)})^{-2} \big) \\ &= \frac{Q^2}{8} - \frac{P_i^2}{2} + 2n^2. \end{aligned} \tag{3.5}$$

For the NS edge, Eqs. (1.3) and (3.1) therefore give

$$h_{1,n}^{(1)} + h_{1,n}^{(2)} = h_1 + 2n^2. \tag{3.6}$$

For a Ramond edge, Eqs. (1.2), (1.4), and (3.1) give

$$h_{i,n}^{(1)} + h_{i,n}^{(2)} = h_i - \frac{1}{16} + 2n^2, \qquad i = 2, 3. \tag{3.7}$$

3.2 The branching theorem

For generic b and generic internal momenta, the tensor-product modules decompose as follows [1, 2]:

$$\mathcal{F}_{\mathrm{NS}} \otimes \mathcal{M}_{\mathrm{NS}}(h_1) \cong \bigoplus_{n\in\frac{1}{2}\mathbb{Z}} \mathcal{V}_{c^{(1)}}(h^{(1)}_{1,n}) \otimes \mathcal{V}_{c^{(2)}}(h^{(2)}_{1,n}), \tag{3.8}$$

$$\mathcal{F}_{\mathrm{R}} \otimes \mathcal{M}_{\mathrm{R}}(\beta_i) \cong \bigoplus_{\epsilon=0,1} \bigoplus_{n\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}} \mathcal{V}_{c^{(1)}}(h^{(1)}_{i,n}) \otimes \mathcal{V}_{c^{(2)}}(h^{(2)}_{i,n}). \tag{3.9}$$

Here $\mathcal{F}$ denotes the auxiliary fermion module, $\mathcal{M}$ the super-Virasoro module, and $\mathcal{V}$ an ordinary Virasoro Verma module. The NS summand labeled by n has fermion parity $2n$ modulo two. The label ϵ in Eq. (3.9) is the parity of the whole Virasoro summand. The two values of ϵ are two distinct copies with the same pair of Virasoro weights.

For clarity, the logical inputs to Eqs. (3.8) and (3.9) are the following.

First, Eq. (2.8) provides an action of the two commuting Virasoro algebras. Second, the free-field construction in Refs. [1, 2] constructs one simultaneous highest-weight state for each $n \in \frac{1}{2}\mathbb{Z}$ in the NS sector and two, of opposite parity, for each $n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$ in the Ramond sector. Their weights are Eq. (3.3). Third, for generic parameters the resulting Virasoro Verma modules are irreducible and have distinct highest weights, except for the deliberate parity multiplicity in the Ramond sector. They are consequently linearly independent. It remains only to show that no states are left over. This follows from the characters, as we now verify.

In the NS sector, the character of the tensor-product module, with its ground-state power suppressed, is

$$\frac{\displaystyle\prod_{k=1}^{\infty}(1+q^{k-1/2})^2}{\displaystyle\prod_{k=1}^{\infty}(1-q^k)}. \tag{3.10}$$

The character of the right side of Eq. (3.8) is

$$\sum_{n\in\frac{1}{2}\mathbb{Z}} \frac{q^{2n^2}}{\displaystyle\prod_{k=1}^{\infty}(1-q^k)^2}. \tag{3.11}$$

The equality of Eqs. (3.10) and (3.11) is exactly the Jacobi identity

$$\sum_{m\in\mathbb{Z}} q^{m^2/2} = \prod_{k=1}^{\infty}(1-q^k)(1+q^{k-1/2})^2, \qquad m = 2n. \tag{3.12}$$

In the Ramond sector the two ground states of each factor give four tensor ground states. The tensor-product character is

$$4\frac{\prod_{k=1}^{\infty}(1+q^k)^2}{\prod_{k=1}^{\infty}(1-q^k)}. \tag{3.13}$$

The character of the right side of Eq. (3.9), including the two parity copies, is

$$2\sum_{n\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}}\frac{q^{2n^2-1/8}}{\prod_{k=1}^{\infty}(1-q^k)^2}. \tag{3.14}$$

Writing $n = k/2 + 1/4$ gives $2n^2 - 1/8 = k(k+1)/2$. Equality of Eqs. (3.13) and (3.14) is therefore the Jacobi identity

$$\sum_{k\in\mathbb{Z}} q^{k(k+1)/2} = 2\prod_{j=1}^{\infty}(1-q^j)(1+q^j)^2. \tag{3.15}$$

Thus the explicitly constructed submodules exhaust the tensor-product modules. This completes the representation-theoretic justification of the branching formulas.

3.3 Ramond ground-state diagonalization

The first Ramond branching labels are $n = \pm 1/4$. We can see both labels and both parity copies directly in the conventions of the main notes. Define only in this subsection

$$\widetilde{w}^- = e^{-i\pi/4}w^-. \tag{3.16}$$

Equations (1.5) become

$$G_0w^+ = i\beta\widetilde{w}^-, \qquad G_0\widetilde{w}^- = i\beta w^+, \qquad \langle\widetilde{w}^-|\widetilde{w}^-\rangle = 1. \tag{3.17}$$

On the four tensor ground states, the only nonscalar term in $L_0^{(1)}$ and $L_0^{(2)}$ is $U_0 = f_0G_0$. Using the graded action in Eq. (2.6), one finds in both parity sectors

$$\begin{aligned}
U_0(u^+\otimes w^+) &= \frac{i\beta}{\sqrt{2}}u^-\otimes\widetilde{w}^-, &\qquad U_0(u^-\otimes\widetilde{w}^-) &= -\frac{i\beta}{\sqrt{2}}u^+\otimes w^+,\\
U_0(u^+\otimes\widetilde{w}^-) &= \frac{i\beta}{\sqrt{2}}u^-\otimes w^+, &\qquad U_0(u^-\otimes w^+) &= -\frac{i\beta}{\sqrt{2}}u^+\otimes\widetilde{w}^-.
\end{aligned} \tag{3.18}$$

It follows without any further assumption that the normalized eigenvectors are

$$v_\pm^0 = \frac{1}{\sqrt{2}}\left(u^+ \otimes w^+ \pm i\, u^- \otimes \widetilde{w}^-\right),$$
$$v_\pm^1 = \frac{1}{\sqrt{2}}\left(u^+ \otimes \widetilde{w}^- \pm i\, u^- \otimes w^+\right), \tag{3.19}$$

where the superscript is the total parity. Their U_0 eigenvalues are $\pm\beta/\sqrt{2}$, respectively. For either parity, call the first and second tensor states in Eq. (3.19) a and d. Equation (3.18) says

$$U_0(a \pm id) = \pm\frac{\beta}{\sqrt{2}}(a \pm id). \tag{3.20}$$

Moreover $\langle a|a\rangle = 1$, $\langle d|d\rangle = -1$, and the mixed pairings vanish. Since the BPZ form is bilinear,

$$\langle v_\pm^\epsilon | v_\pm^\epsilon\rangle = \frac{1}{2}\big(1 + (\pm i)^2(-1)\big) = 1, \qquad \langle v_+^\epsilon | v_-^\epsilon\rangle = \frac{1}{2}\big(1 + i(-i)(-1)\big) = 0. \tag{3.21}$$

States of different parity are also orthogonal.

Substitution into Eq. (2.8) shows that v_+^ϵ has the pair of weights labeled by $n = -1/4$ and v_-^ϵ the pair labeled by $n = +1/4$, for the correlated branch choice in Eq. (3.4). To check the sign of this assignment, note that the term in $h_{i,n}^{(1)}$ which is odd in n is $-2nP_i b^{(1)}/\sqrt{2-2b^2}$. At $n = -1/4$ it equals $bP_i/(2(1-b^2))$, which is exactly the $U_0 = +\beta_i/\sqrt{2} = +P_i/2$ term in $L_0^{(1)}$. Thus the plus eigenvector indeed has label $-1/4$; the minus case follows by changing both signs. Equation (3.7) gives

$$h_{i,\pm 1/4}^{(1)} + h_{i,\pm 1/4}^{(2)} = h_i + \frac{1}{16}, \tag{3.22}$$

which is exactly the sum of the super-Virasoro Ramond ground weight and the auxiliary fermion Ramond ground weight. Thus the four states in Eq. (3.19) account for all four tensor ground states and explicitly verify the lowest part of Eq. (3.9) with the phases of the main notes.

4 Restriction of the three-point forms

4.1 Branch highest states

Choose simultaneous Virasoro highest states

$$v_{1,n_1}, \qquad v_{2,n_2}^{\epsilon_2}, \qquad v_{3,n_3}^{\epsilon_3} \tag{4.1}$$

in the summands of Eqs. (3.8) and (3.9). Thus

$$n_1 \in \frac{1}{2}\mathbb{Z}, \qquad n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}, \qquad \epsilon_2, \epsilon_3 \in \{0, 1\}. \tag{4.2}$$

We normalize every state in Eq. (4.1) to have BPZ norm one. This is possible for generic parameters. The remaining sign of each state will cancel between the two theta vertices.

The parity of v_{1,n_1} is $2n_1$ modulo two, while the parities of the other two states are ϵ_2 and ϵ_3. Since the two Virasoro algebras are even subalgebras, every descendant in one of these three Virasoro modules has the same total fermion parity as its highest state.

4.2 The product three-point form

Let $\rho_f^{(\eta)}$ be the stripped super-Virasoro NS–R–R form used in the main notes. Its parity is f and its Ramond chiral-structure label is η. Let ρ_g^{f} be a fixed stripped NS–R–R form of the auxiliary fermion, of parity g. Its ground-state normalization is held fixed throughout. If one wishes to keep more than one auxiliary-fermion spin-field structure, the argument below applies to each such form separately.

There are two oriented product forms because the order of the two tensor factors at the second pair of pants is opposite to their order at the first. We derive both signs. For three homogeneous decomposable states, put

$$\mu_i = |u_i|, \qquad \nu_i = |x_i|, \qquad p_i = \mu_i + \nu_i \pmod 2. \tag{4.3}$$

At the left vertex the local order is $u_1, x_1, u_2, x_2, u_3, x_3$. Moving x_1 through u_2, u_3 and then moving x_2 through u_3 gives

$$\begin{aligned} &R_{f,g}^{(\eta),L}(u_1 \otimes x_1, u_2 \otimes x_2, u_3 \otimes x_3) \\ &\quad = (-1)^{\nu_1(\mu_2+\mu_3)+\nu_2\mu_3} \rho_g^{\mathrm{f}}(u_1, u_2, u_3) \rho_f^{(\eta)}(x_1, x_2, x_3). \end{aligned} \tag{4.4}$$

At the right vertex the BPZ-oriented local order is $x_1, u_1, x_2, u_2, x_3, u_3$. Moving u_1 through x_2, x_3 and then moving u_2 through x_3 gives

$$\begin{aligned} &R_{f,g}^{(\eta'),R}(u_1 \otimes x_1, u_2 \otimes x_2, u_3 \otimes x_3) \\ &\quad = (-1)^{\mu_1(\nu_2+\nu_3)+\mu_2\nu_3} \rho_g^{\mathrm{f}}(u_1, u_2, u_3) \rho_f^{(\eta')}(x_1, x_2, x_3). \end{aligned} \tag{4.5}$$

Both forms have parity $f + g$.

The distinction between Eqs. (4.4) and (4.5) is essential. Direct expansion gives

$$\begin{aligned} &\sum_{i<j} p_i p_j + \nu_1(\mu_2 + \mu_3) + \nu_2\mu_3 + \mu_1(\nu_2 + \nu_3) + \mu_2\nu_3 \\ &\qquad\qquad = \sum_{i<j} \mu_i\mu_j + \sum_{i<j} \nu_i\nu_j \pmod 2. \end{aligned} \tag{4.6}$$

Indeed, expanding $\sum_{i<j} p_i p_j$ gives the right side of Eq. (4.6) together with six mixed monomials. The left-vertex exponent is the sum of the three monomials $\nu_1\mu_2, \nu_1\mu_3, \nu_2\mu_3$; the right-vertex

exponent is the sum of the remaining three. Every mixed monomial therefore occurs twice and vanishes modulo two. The first term is the theta sewing sign of the three tensor-product states. The next two terms are the vertex signs. The right side is the sum of the separate theta sewing exponents of the auxiliary-fermion and super-Virasoro blocks. Equation (4.6) is therefore the complete term-by-term sign check for tensor-product factorization. The tensor-product Gram matrix introduces no additional sign because its bilinear form is the ordinary product of the two bilinear forms. In particular, no residual factor $(-1)^{fg}$ remains.

The oriented branching coefficients are

$$\begin{aligned} &B^{(\eta),L}_{f,g}(n_1, n_2, n_3; \epsilon_2, \epsilon_3) \\ &\qquad = R^{(\eta),L}_{f,g}(v_{1,n_1}, v^{\epsilon_2}_{2,n_2}, v^{\epsilon_3}_{3,n_3}), \end{aligned} \tag{4.7}$$

$$\begin{aligned} &B^{(\eta'),R}_{f,g}(n_1, n_2, n_3; \epsilon_2, \epsilon_3) \\ &\qquad = R^{(\eta'),R}_{f,g}(v_{1,n_1}, v^{\epsilon_2}_{2,n_2}, v^{\epsilon_3}_{3,n_3}). \end{aligned} \tag{4.8}$$

There is no universal relation of the form $B^R = (-1)^{p_1p_2+p_1p_3+p_2p_3} B^L$: that would identify the two different vertex reorderings. There is also no complex conjugation between the coefficients because the BPZ forms are bilinear.

Parity alone gives the selection rule

$$B^{(\eta),L}_{f,g} = 0 = B^{(\eta'),R}_{f,g} \quad \text{unless} \quad 2n_1 + \epsilon_2 + \epsilon_3 = f + g \pmod 2. \tag{4.9}$$

The arguments $(n_1, n_2, n_3; \epsilon_2, \epsilon_3)$ are suppressed in this equation. To see the rule directly, a form of parity $f + g$ can be nonzero only when the sum of its three input parities equals $f + g$; the three branch parities are $2n_1, \epsilon_2, \epsilon_3$.

We next determine all descendant matrix elements from Eq. (4.7). Restrict $R^{(\eta),L}_{f,g}$ to the three chosen $\mathrm{Vir}_{c^{(1)}} \oplus \mathrm{Vir}_{c^{(2)}}$ summands. Apply the Ward identities of the first Virasoro algebra while holding the second Virasoro descendants fixed. For generic weights, the Virasoro three-point form is unique after its value on the three highest states is specified. Hence the dependence on all first-copy descendants is the standard stripped Virasoro three-point form. Applying the Ward identities of the second, commuting Virasoro algebra gives the same statement for the second-copy descendants. Therefore, for arbitrary partitions A_i, C_i,

$$\begin{aligned} &R^{(\eta),L}_{f,g}\left(L^{(1)}_{-A_1}L^{(2)}_{-C_1}v_{1,n_1}, L^{(1)}_{-A_2}L^{(2)}_{-C_2}v^{\epsilon_2}_{2,n_2}, L^{(1)}_{-A_3}L^{(2)}_{-C_3}v^{\epsilon_3}_{3,n_3}\right) \\ &\quad = B^{(\eta),L}_{f,g}(n_1, n_2, n_3; \epsilon_2, \epsilon_3) \\ &\qquad \times \rho^{(1)}(L^{(1)}_{-A_1}v_{1,n_1}, L^{(1)}_{-A_2}v^{\epsilon_2}_{2,n_2}, L^{(1)}_{-A_3}v^{\epsilon_3}_{3,n_3}) \\ &\qquad \times \rho^{(2)}(L^{(2)}_{-C_1}v_{1,n_1}, L^{(2)}_{-C_2}v^{\epsilon_2}_{2,n_2}, L^{(2)}_{-C_3}v^{\epsilon_3}_{3,n_3}). \end{aligned} \tag{4.10}$$

Both ordinary Virasoro forms on the right are normalized to one on the three branch highest states. Equation (4.10) is the exact point at which the branching coefficient enters. No descendant-dependent coefficient remains. The same Ward-identity argument at the oriented right vertex gives the identical factorization with $B^{(\eta'),R}_{f,g}$ in place of $B^{(\eta),L}_{f,g}$.

5 Finite-product form of the branching coefficients

The algebra embedding determines the descendant dependence but not the restriction to three branch highest states. We now state exactly what is known about that restriction. The finite products below are unambiguous algebraic expressions. Their identification with the general mixed Ramond branching coefficient is conditional, as explained in Subsection 5.3.

5.1 One parity-sensitive product

For a nonnegative integer m, define

$$\ell(x,m) = \prod_{\substack{r,s\in\mathbb{Z}_{\geq 0},\ r+s<m \\ r+s\equiv m \pmod 2}} \left(x + rb + sb^{-1}\right). \tag{5.1}$$

The product is one when its index set is empty. In particular,

$$\ell(x,0) = \ell(x,1) = 1, \qquad \ell(x,2) = x, \qquad \ell(x,3) = (x+b)(x+b^{-1}). \tag{5.2}$$

Changing m to $m+2$ leaves all old lattice points and adds precisely the diagonal $r+s=m$. Therefore

$$\ell(x,m+2) = \ell(x,m)\prod_{r=0}^{m}\left(x + rb + (m-r)b^{-1}\right). \tag{5.3}$$

This proves the recurrence directly from the definition.

The number of factors in Eq. (5.1) is $\lfloor m^2/4\rfloor$. For $m=2a$ the allowed diagonal lengths sum to $1+3+\cdots+(2a-1)=a^2$; for $m=2a+1$ they sum to $2+4+\cdots+2a=a(a+1)$. For $m>0$ define

$$\ell(x,-m) = (-1)^{\lfloor m^2/4\rfloor}\ell(Q-x,m). \tag{5.4}$$

Applying Eq. (5.4) twice returns $\ell(x,m)$, so this continuation is an involution. For even $m=2a$ its sign is $(-1)^a$; for odd m it is one. These are the standard even- and odd-screening reflection rules.

5.2 The candidate

Every second argument in the following expression is an integer because of Eq. (4.2). Define

$$\mathcal{C}(n_1,n_2,n_3) = \frac{\begin{gathered}\ell\Big(\frac{Q}{2}+P_1+P_2+P_3,2(n_1+n_2+n_3)\Big)\,\ell\Big(\frac{Q}{2}+P_1+P_2-P_3,2(n_1+n_2-n_3)\Big)\\ \times\,\ell\Big(\frac{Q}{2}+P_1-P_2+P_3,2(n_1-n_2+n_3)\Big)\,\ell\Big(\frac{Q}{2}-P_1+P_2+P_3,2(-n_1+n_2+n_3)\Big)\end{gathered}}{\sqrt{\prod_{i=1}^{3}\ell(2P_i,4n_i)\ell(Q+2P_i,4n_i)}}. \tag{5.5}$$

After fixing the auxiliary-fermion ground form and the signs of the branch highest states, the finite-product candidate asserts that

$$\begin{aligned} B^{(\eta),L}_{f,g} &= \kappa^{(\eta),L}_{f,g}\,\mathcal{C}(n_1,n_2,n_3),\\ B^{(\eta'),R}_{f,g} &= \kappa^{(\eta'),R}_{f,g}\,\mathcal{C}(n_1,n_2,n_3), \end{aligned} \tag{5.6}$$

when Eq. (4.9) holds; both coefficients vanish otherwise. The arguments $(n_1,n_2,n_3;\epsilon_2,\epsilon_3)$ of the two κ factors are suppressed. These factors are independent of the continuous momenta but may depend on all displayed discrete labels. They contain the chosen auxiliary-fermion ground normalization and the finite Ramond zero-mode change of basis. They are not unrecorded Koszul signs: every Koszul sign needed for sewing is already displayed in Eq. (4.6). Equivalently, for fixed discrete labels one may define κ as the quotient of B by $\mathcal{C}$; the nontrivial assertion in Eq. (5.6) is that this quotient has no continuous-momentum dependence. A different choice of square-root branches in Eq. (5.5) is absorbed into κ.

The four screening parities in Eq. (5.5) are forced by the branch lattices. The first second argument differs from the second by $4n_3$, from the third by $4n_2$, and from the fourth by $4n_1$. The first two differences are odd and the last is even. Thus the first and fourth factors have the same screening parity, while the second and third have the opposite parity. Exactly two factors use the even lattice and two use the odd lattice. On the legs, $4n_1$ is even and $4n_2, 4n_3$ are odd, as required for one NS and two Ramond modules.

5.3 Proof status

When all three modules are NS, the four numerator factors and the three leg normalizations in Eq. (5.5) are the normalized blow-up factor proved in Ref. [3]. Their notation has $\alpha_i = Q/2+P_i$; hence their four arguments $\alpha_1+\alpha_2+\alpha_3-Q$, $\alpha_1+\alpha_2-\alpha_3$, $\alpha_1+\alpha_3-\alpha_2$, and $\alpha_2+\alpha_3-\alpha_1$ are exactly the four arguments in Eq. (5.5). Their branch label is $2n_i$. Consequently their four product lengths are the four linear combinations of the $2n_i$ displayed there, while unit normalization supplies the three leg lengths $4n_i$. Their remaining powers of two and branch-state phases are independent of the continuous momenta and belong to the corresponding κ factors.

For the mixed NS–R–R sector, Ref. [4] proposed the double-Liouville correspondence for half-integral branch labels and checked several nontrivial three-point functions. It did not prove the general mixed three-point formula. Nor is there a proved Ramond analogue of the one-leg normalization used in the NS blow-up factor. The closest published closed formula is the conjectured Ramond Whittaker coefficient of Ref. [2]. In its notation, for $n > 0$,

$$s_{\text{odd}}(x, 2n) = 2^{1/8}\ell(x, 4n). \tag{5.7}$$

The denominator of that conjectured coefficient is therefore, apart from the fixed factor $2^{1/4}$,

$$\ell(2P, 4n)\ell(Q + 2P, 4n), \tag{5.8}$$

which is the Ramond leg factor in Eq. (5.5). The Whittaker coefficient also contains fixed Whittaker-coordinate rescalings, so this agreement is evidence for the leg factor, not a proof of a BPZ norm formula in the present conventions. The conjecture was checked in Ref. [2] through $|n| = 9/4$ but was not proved.

Consequently Eq. (5.6) is a conditional formula in the mixed sector. It follows if the general Ramond double-Liouville chiral correspondence and the one-leg normalization in Eq. (5.8) are assumed in the present normalization. Neither assumption is needed for the exact block identity below.

There is one unconditional mixed-sector check. For the Ramond ground branches $n = \pm 1/4$, Eqs. (5.2) and (5.4) give $\ell(x, 4n) = 1$. If $n_1 = 0$ and $n_2, n_3 = \pm 1/4$, all four numerator labels in Eq. (5.5) are 0 or ± 1. Hence

$$\mathcal{C}(0, n_2, n_3) = 1. \tag{5.9}$$

This agrees with the explicit normalized states in Eq. (3.19): at ground level all remaining information is a finite zero-mode matrix element contained in κ.

6 The theta-channel block identity

6.1 Factorization before branching

Let $\mathbb{F}^{\mathrm{f}}_g$ be the auxiliary-fermion theta block obtained by sewing two copies of ρ^{f}_g with the same plumbing parameters (q_i, η_i) as the superconformal block. Strip its ground propagator on every edge, so its expansion starts at total plumbing level zero. Likewise let $\mathbb{F}^{\mathrm{tot},(\eta,\eta')}_{f,g}$ be the block obtained by the canonical graded tensor product of the super-Virasoro and auxiliary-fermion sewings. Equivalently, its two oriented pair-of-pants forms are the forms used in Eqs. (4.4) and (4.5). The tensor-product bilinear form and its Gram matrix are the ordinary products of the two factor forms and Gram matrices.

Consider one homogeneous term. Its three total parities are the p_i in Eq. (4.3). Its total theta sewing sign is $(-1)^{\sum_{i<j} p_i p_j}$. Multiplication by the two vertex signs in Eqs. (4.4) and (4.5) gives, by Eq. (4.6), the product of the separate sewing signs $(-1)^{\sum_{i<j} \mu_i \mu_j}$ and $(-1)^{\sum_{i<j} \nu_i \nu_j}$. The inverse Gram matrices introduce no sign. The powers of q_i add, and $\eta_i^{\mu_i} \eta_i^{\nu_i} = \eta_i^{p_i}$ because $\eta_i^2 = 1$. Thus every homogeneous summand agrees on the two sides of

$$\mathbb{F}^{\mathrm{tot},(\eta,\eta')}_{f,g} = \mathbb{F}^{(\eta,\eta')}_{f} \mathbb{F}^{\mathrm{f}}_{g}. \tag{6.1}$$

This is a factorization of the block of the direct-sum algebra. It does not yet use the two Virasoro subalgebras.

6.2 Resolution of the propagators

Insert the decompositions (3.8) and (3.9) into the three propagators of the total block. Different branching summands are orthogonal for generic parameters: summands of different weight are orthogonal by $L_0^{(i)}$ invariance, and the two Ramond copies of equal weight are orthogonal because they have opposite parity. Thus a branch label chosen at one end of a tube must be the same at the other end.

Within one branching summand, the Gram matrix is a tensor product of the two ordinary Virasoro Gram matrices. Equations (4.10) at the two vertices then show that the descendant sewing sum is exactly a product of two ordinary theta blocks. Denote the ordinary stripped block by

$$F_\theta(c; h_1, h_2, h_3). \tag{6.2}$$

It contains the descendant powers of the q_i but not the three primary propagators.

It remains to determine the powers preceding these two ordinary blocks. On the NS edge, Eq. (3.6) shows that the branch highest state occurs at level $2n_1^2$ above the tensor ground state. On a Ramond edge, the tensor ground weight is $h_i + 1/16$, so Eq. (3.7) shows that the branch highest state occurs at level

$$\left(h_i - \frac{1}{16} + 2n_i^2\right) - \left(h_i + \frac{1}{16}\right) = 2n_i^2 - \frac{1}{8}. \tag{6.3}$$

Finally, the lift η_i^F acts on an entire branching summand by the parity of its highest state because both Virasoro algebras are even. The three lift factors are therefore

$$\eta_1^{2n_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3}. \tag{6.4}$$

6.3 Exact identity

Combining Eqs. (6.1)– (6.4) gives

$$
\begin{aligned}
&\mathbb{F}_f^{(\eta,\eta')}(h_1,\beta_2,\beta_3,c)\mathbb{F}_g^{\mathrm{f}} \\
=&\sum_{\substack{n_1\in\frac{1}{2}\mathbb{Z},\quad n_2,n_3\in\frac{1}{2}\mathbb{Z}+\frac{1}{4}\\ \epsilon_2,\epsilon_3\in\{0,1\}\\ 2n_1+\epsilon_2+\epsilon_3=f+g \pmod 2}} q_1^{2n_1^2}q_2^{2n_2^2-1/8}q_3^{2n_3^2-1/8}\eta_1^{2n_1}\eta_2^{\epsilon_2}\eta_3^{\epsilon_3} \\
&\quad\times(-1)^{(2n_1)\epsilon_2+(2n_1)\epsilon_3+\epsilon_2\epsilon_3} \\
&\quad\times B_{f,g}^{(\eta),L}(n_1,n_2,n_3;\epsilon_2,\epsilon_3)B_{f,g}^{(\eta'),R}(n_1,n_2,n_3;\epsilon_2,\epsilon_3) \\
&\quad\times F_\theta\left(c^{(1)};h_{1,n_1}^{(1)},h_{2,n_2}^{(1)},h_{3,n_3}^{(1)}\right) \\
&\quad\times F_\theta\left(c^{(2)};h_{1,n_1}^{(2)},h_{2,n_2}^{(2)},h_{3,n_3}^{(2)}\right).
\end{aligned}
\tag{6.5}
$$

Every symbol on the right side has now been defined. In particular, $c^{(i)}$ is given by Eqs. (2.18) and (2.19), the six Virasoro weights are given by Eqs. (3.1)–(3.3), and the two oriented branching coefficients are defined in Eqs. (4.7) and (4.8).

If the conditional formula (5.6) is used, the coefficient multiplying the two ordinary blocks becomes

$$
\begin{aligned}
&(-1)^{(2n_1)\epsilon_2+(2n_1)\epsilon_3+\epsilon_2\epsilon_3}B_{f,g}^{(\eta),L}B_{f,g}^{(\eta'),R} \\
&\quad=(-1)^{(2n_1)\epsilon_2+(2n_1)\epsilon_3+\epsilon_2\epsilon_3}\kappa_{f,g}^{(\eta),L}\kappa_{f,g}^{(\eta'),R}\mathcal{C}(n_1,n_2,n_3)^2.
\end{aligned}
\tag{6.6}
$$

Changing the sign of any one branch highest state changes both oriented coefficients and therefore leaves their product invariant. Accordingly, Eq. (6.6) contains no square-root ambiguity.

If $\mathbb{F}_g^{\mathrm{f}}$ has a nonzero constant term, it is invertible as a formal plumbing series. Dividing Eq. (6.5) then gives the desired expression for the individual Ramond block:

$$
\boxed{\mathbb{F}_f^{(\eta,\eta')}=\frac{1}{\mathbb{F}_g^{\mathrm{f}}}\,(\text{right side of Eq. (6.5)})\,.}
\tag{6.7}
$$

For a fermion spin structure with an unsaturated zero mode, $\mathbb{F}_g^{\mathrm{f}}$ can vanish. Equation (6.5) is still true, but it cannot then be divided to obtain the superconformal block. One must instead choose an auxiliary fermion form with the zero mode soaked, or combine several component identities before solving for the desired superconformal component.

7 Checks and consequences

7.1 Local finiteness

The identity is an infinite branching sum, but it is locally finite as a formal series in the plumbing parameters. Indeed, a contribution to the coefficient of $q_1^{N_1} q_2^{N_2} q_3^{N_3}$ must obey

$$2n_1^2 \leq N_1, \qquad 2n_2^2 - \frac{1}{8} \leq N_2, \qquad 2n_3^2 - \frac{1}{8} \leq N_3. \tag{7.1}$$

Only finitely many half-integral or quarter-shifted labels satisfy these bounds. The first Ramond labels and their levels are

n	$-5/4$	$-3/4$	$-1/4$	$1/4$	$3/4$	$5/4$
$2n^2 - 1/8$	3	1	0	0	1	3

Thus Eq. (6.5) can be evaluated to any fixed plumbing order using only finitely many ordinary Virasoro blocks.

7.2 The ground term

At $q_1 = q_2 = q_3 = 0$, only

$$n_1 = 0, \qquad n_2, n_3 \in \left\{ -\frac{1}{4}, +\frac{1}{4} \right\} \tag{7.2}$$

survive. Both ordinary stripped Virasoro blocks have constant term one. Equation (6.5) reduces to the finite identity

$$\begin{aligned}
&\mathbb{F}_f^{(\eta,\eta')}\Big|_{q_i=0} \mathbb{F}_g^{\mathrm{f}}\Big|_{q_i=0} \\
&= \sum_{\substack{n_2, n_3 = \pm 1/4 \\ \epsilon_2 + \epsilon_3 = f + g \pmod 2}} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3} (-1)^{\epsilon_2 \epsilon_3} B_{f,g}^{(\eta),L}(0, n_2, n_3; \epsilon_2, \epsilon_3) B_{f,g}^{(\eta'),R}(0, n_2, n_3; \epsilon_2, \epsilon_3).
\end{aligned} \tag{7.3}$$

The superconformal factor on the left is already known in the conventions of the main notes:

$$\begin{aligned}
\mathbb{F}_0^{(\eta,\eta')}\Big|_{q_i=0} &= 1 + \eta\eta'\eta_2\eta_3, \\
\mathbb{F}_1^{(\eta,\eta')}\Big|_{q_i=0} &= i(\eta\eta'\eta_2 - \eta_3).
\end{aligned} \tag{7.4}$$

Therefore Eq. (7.3) is an immediate and phase-sensitive normalization check on any explicit computation of the branching coefficients. The four states in Eq. (3.19) make this check a finite 4×4 ground-state calculation; no descendant Ward identities are needed.

7.3 Status of the branching coefficients

The following statements are exact. The branching coefficients are the primary restrictions in Eqs. (4.7) and (4.8); all descendant matrix elements then follow from the two ordinary Virasoro Ward identities through Eq. (4.10). The coefficients are independent of the plumbing parameters, they vanish unless the parity rule (4.9) holds, and only finitely many enter at any fixed plumbing order. The left and right coefficients belong to two different orderings of the product three-point form, so there is no universal sign relating them.

For three NS modules, the continuous-momentum dependence and the three leg normalizations are given by the finite product $\mathcal{C}$ in Eq. (5.5). This is the normalized blow-up factor proved in Ref. [3]. Powers of two and choices of branch-state phases are independent of the continuous momenta and are contained in the corresponding κ factors.

For one NS and two Ramond modules, three pieces of evidence support the same finite product. The proposed Ramond double-Liouville correspondence reproduces nontrivial mixed three-point functions. The conjectured Ramond Whittaker coefficient gives the leg factor (5.8) and was checked through $|n| = 9/4$. Finally, the product gives $\mathcal{C} = 1$ on every Ramond ground branch, in agreement with the explicit normalized ground states. These checks do not prove the general mixed formula. Thus Eq. (5.6) remains conditional in the NS–R–R sector.

The undetermined data are discrete. They are the zero-mode and auxiliary-fermion normalization factors κ for the chosen chiral structures and branch parities. At any fixed branch level they can be found by a finite algebraic calculation: construct the branch highest states, normalize their BPZ forms, and evaluate Eqs. (4.4) and (4.5). No further Grassmann sign may be inserted after this evaluation; the complete sewing sign is already displayed in Eq. (6.5). Neither the conditional product formula nor the determination of κ is required for the exact identity, which remains valid with the branching coefficients defined directly by the three-point forms.

7.4 Generic parameters and continuation

The branching theorem was stated for generic parameters so that all Verma modules are irreducible, their norms are nonzero, and distinct summands do not collide. Both sides of Eq. (6.5) are meromorphic functions of the central charge and the momenta at every fixed plumbing order. The identity therefore extends by meromorphic continuation away from its poles. At a degenerate value one may either take this limit or replace the Verma modules by the appropriate irreducible quotients. The points $b^2 = 1$, at which the displayed generators have singular coefficients, are likewise reached only after taking a limit of the generic identity.

8 Conclusion

The auxiliary Majorana fermion converts the mixed NS–R–R super-Virasoro problem into a branching problem for two commuting ordinary Virasoro algebras. The Ramond sector is not an obstruction, but it changes the branching lattice from $\frac{1}{2}\mathbb{Z}$ to $\frac{1}{2}\mathbb{Z}+\frac{1}{4}$ and produces two parity copies of every Virasoro pair. These facts give the powers $q^{2n^2-1/8}$ and the parity selection rule in Eq. (6.5). The left vertex sign, the right vertex sign, and the propagator sign obey the elementary identity (4.6); this is why the product with the auxiliary-fermion block factorizes without a hidden phase. The exact identity expresses that product as a locally finite sum of products of ordinary theta blocks. Equation (5.5) gives the natural closed finite-product form of the momentum dependence. Its all-NS version is proved, whereas its mixed Ramond version and the general Ramond one-leg normalization remain conditional. The only remaining convention-dependent data are the finite zero-mode factors κ.

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